on the retina. When we see a tower, its image appears on the retina with the top of the tower pointing downward. When this phenomenon was established, a number of scientists tried to detect in the human organ of sight that particular mechanism which supposedly re-inverts the image on the retina. This is a phantom problem, and never can be anything else, for it is based on an erroneous premise, for which there can be no possible proof—namely, that in the organ of vision the image of an object must be upright rather than inverted.

Far more difficult than those cases in which, as in the examples just cited, the assumptions are mistaken, are problems whose presuppositions contain no error, but are so vaguely worded that they must remain phantom ones because they are inadequately formulated. Yet, it so happens that it is just such cases with which we shall be chiefly preoccupied.

My first example is a phantom problem, for the triviality of which I beg your forgiveness. The room in which we now sit, has two side walls, a right-hand one and a left-hand one. To you, *this* is the right side, to me, sitting facing you, *that* is the right side. The problem is: Which side is in reality the right-hand one? I know that this question sounds ridiculous, yet I dare call it typical of an entire host of problems which have been, and in part still are, the subject of earnest and learned debates, except that the situation is not always quite so clear. It demonstrates, right at the very outset, what great caution must be exercised in using the word, *real.* In many instances, the word has any sense at all only when the speaker first defines clearly the point of view on which his considerations are based. Otherwise, the words, *real* or *reality,* are often empty and misleading.

Another example: I see a star shining in the sky. What is *real* in it? Is it the glowing substance, of which it is composed, or is it the sensation of light in my eyes? The question is meaningless so long as I do not state whether I am assuming a realistic or a positivistic point of view.

Still another example, this one from the realm of modern physics: When the behavior of a moving electron is studied through an electron microscope, the electron appears as a particle following a definite course. But when the electron is made to pass through a crystal, the image projected on the screen shows every characteristic of a refracted light wave. The question, whether the electron is in reality a particle, occupying a certain position in space at a certain time, or a wave, filling all of infinite space, will therefore constitute a phantom problem so long as we fail to stipulate which of the two viewpoints is applied in the study of the behavior of the electron.

The famous controversy between Newton’s emission theory and Huygens’ wave theory of light is also a phantom problem of science. For every decision for or against either of these two opposing theories will be a completely arbitrary one, depending on whether one accepts the point of view of the